Srijith Sai Kurelli

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.33
Fiitjee Junior College	Class XII (Telangana Board of Intermediate Education)	2024	98.60%
Delhi Public School Nacharam	Class X (CBSE)	2022	98.80%

SCHOLASTIC ACHIEVEMENTS

- Currently **ranked 1st** in the Engineering Science department at IIT Hyderabad, with a **peak SGPA of 9.83** in Semester 2 and maintaining a **9+ SGPA** across three consecutive semesters.
- Achieved a peak **LeetCode rating** of **1687** through consistent participation in coding contests and algorithmic problem solving.
- Awarded **Gold Honour** in the **International Youth Math Challenge (IYMC)** 2023, placing among the **top 3%** of participants worldwide.
- Secured **Zonal Rank 9** in Telangana in the **SOF International Mathematics Olympiad** 2020-21.

PROJECTS

Movie Dialogue Generation | *Python, TensorFlow, Keras, Git, NLP* | GitHub Link

- Developed an end-to-end LSTM-based NLP pipeline for movie dialogue generation using the Cornell Movie Dialog Corpus, generating **2.9M+ training sequences** from **304K+ processed dialogues.**
- Designed and evaluated **baseline and enhanced LSTM architectures** by increasing LSTM capacity from **128 to 256 units**, incorporating dropout-based regularization and an additional dense layer to improve dialogue coherence.
- Implemented and compared three decoding strategies--**Greedy Search, Top-k Sampling, and Temperature Sampling**--for quantitative and qualitative evaluation of generated dialogue.

Credit Card Fraud Detection | *Python, Scikit-learn, XGBoost, Pandas* | GitHub Link

- Developed a **credit card fraud detection** system using **Random Forest** and **XGBoost** classifiers with comprehensive model evaluation.
- Achieved **99.96% accuracy** and **0.8743 F1-score**, with **Random Forest** outperforming XGBoost across **all evaluation metrics**, including Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Steam Game Recommender | *Python, Pandas, Scikit-learn, XGBoost, Git* | GitHub Link

- Built a hybrid Steam game recommender using **content-based filtering** with **XGBoost re-ranking** for personalized recommendations.
- Developed an end-to-end ML pipeline, generating **503K+ training examples** from **44.5K Steam users**.
- Evaluated **Logistic Regression**, **Random Forest**, and **XGBoost**, achieving a **ROC-AUC of 0.715** with the final model.

URL Shortener API | *Python, FastAPI, PostgreSQL, SQLAlchemy, Git* | GitHub Link

- Built a RESTful URL Shortener API using **FastAPI, PostgreSQL, and SQLAlchemy**, enabling unique 6-character short URL generation, persistent storage, and seamless URL redirection.
- Implemented a modular layered architecture with **Pydantic** validation and **URL analytics** (click counts and last-accessed timestamps), while documenting and testing APIs using **Swagger UI** and **Postman**.

Key-Value Storage Engine | *C++, STL, Multithreading, Git* | GitHub Link

- Developed a **thread-safe in-memory** Key-Value Storage Engine in C++ with a modular Parser-Executor-Storage architecture, supporting concurrent CRUD operations using **threads, mutexes, and condition variables**.
- Implemented **automatic TTL expiration** and **O(1) LRU eviction** using efficient STL data structures with a background cleanup thread.
- Validated thread safety through an **8-thread stress test** executing **80K+ concurrent operations**, with runtime storage statistics.

SKILLS

Programming Languages: C, C++, Python, SQL

Python Libraries: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost

Backend & Databases: FastAPI, PostgreSQL, SQLite

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Postman

Soft Skills: Leadership, Problem Solving, Analytical Thinking, Team Collaboration

RELEVANT COURSES

- **Artificial Intelligence & Machine Learning:** Introduction to AI, Machine Learning Specialization by DeepLearning.AI
- **Computer Science:** Introduction to Programming, Data Structures and Applications, Algorithms, Theory of Computation, Discrete Mathematics for Computer Science, Introduction to Computing, Operating Systems-I
- **Mathematics:** Probability and Random Variables, Introduction to Statistics, Elementary Linear Algebra, Differential Equations, Complex Variables, Calculus I & II

POSITIONS OF RESPONSIBILITY

Head of Operations | Entrepreneurship Cell (E-Cell), IIT Hyderabad

- Leading the Operations vertical, managing a team of 10 Managers and 20+ Assistant Managers across E-Cell.
- Overseeing logistics, vendor negotiations, venue and resource management, stakeholder coordination, and end-to-end operations for flagship events including E-Summit'27 and E-Merge 4.0.

Head Internship Coordinator | Office of Career Services (OCS), IIT Hyderabad

- Representing the Engineering Science department and coordinating internship activities with OCS, recruiters, and students.
- Coordinating internship drives, recruiter interactions, and student communication to ensure seamless recruitment operations.

UG Mentor | Sunshine - The Counselling Cell, IIT Hyderabad

- Mentored first-year students, supporting their academic, social, and personal transition to campus life.
- Facilitated peer support by guiding freshers to campus resources and fostering an inclusive, supportive environment.

EXTRACURRICULAR

- **Interned at Nirmaan Organization** under the **Career Saathi Program**, creating resources and conducting career guidance sessions.
- Interested in word games and regularly solve the daily **Wordle** and **Strands puzzles** by ***The New York Times***.
- Volunteered with the **National Service Scheme**, IIT Hyderabad, contributing to community outreach and social

welfare initiatives.